

2023



Advancing and Accelerating Data Security Using Amazon Redshift with Bank Leumi

Learn how Bank Leumi advanced data security using Amazon Redshift-based cyber data analysis.

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6 minutes to scale

an Amazon Redshift cluster versus months on premises

Minutes instead of hours

to perform forensic analysis

Minutes instead of hours

to materialize table view of structured



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Overview

[Bank Leumi](#) (Leumi), a leading Israeli bank, is committed to keeping its customers' data and money secure. At the core of Leumi's robust IT infrastructure lies forensic analytics—an indispensable tool that is used to identify, probe, and thoroughly document security incidents and financial crimes through meticulous analysis of cyber data. The insights gleaned from this process serve as proactive measures to prevent future occurrences, protecting the safety and integrity of the bank's operations.

Leumi's forensic analytics solution was used in every department in the cyber division, but because it was hosted on premises, the bank found that it was difficult to scale and couldn't support its growing data needs. Leumi saw this challenge as an opportunity to modernize on Amazon Web Services (AWS).

Aided by [AllCloud](#), an [AWS Partner](#), Leumi is migrating its cyber forensic analytics solution to the cloud, improving its detection and response capabilities. The bank's security operations center (SOC) analysts can analyze cyber data faster, gaining crucial security insights to better protect customers' accounts and data. With this cutting-edge approach, Leumi can quickly identify security threats and take action while fostering a safer banking experience.



Opportunity | Using AWS Services to Protect Leumi Customers

Leumi is the oldest banking corporation in Israel and one of the largest, with some 180 branches across Israel and other international locations. The bank hosts massive amounts of data; it hosts a total of 1 PB of cyber data and ingests around 6 TB every single day.



Leumi's security forensics analytics solution processes and analyzes data sourced from hundreds of security and application logs spanning every sector of the bank. These insights empower Leumi to detect and assess security incidents, determine their extent and impact, and gather critical evidence for potential legal or compliance matters.

Without these insights, customers might be more vulnerable—which is why Leumi requires fast access to cyber data and scalable analytics. However, making the most of this data requires capabilities that Leumi's on-premises infrastructure couldn't support. As part of its greater cloud journey, it chose to migrate its cyber forensic analytics solution to AWS.

Leumi engaged AllCloud, a professional services company that specializes in cloud transformation, to perform the migration. "AllCloud and its team of data engineers worked together with Leumi's cyber data team to conduct the migration: changing extract, transform, load processes and updating reports and stored procedures," says Dudi Levi, head of data, cyber division at Leumi.

Solution | Performing a Forensic Analysis in Hours Using Amazon Redshift

To facilitate a safe migration of its legacy data security pipelines, Leumi and AllCloud designed a two-phase cloud migration based on AWS best practices. Beginning in early 2022, the first phase was to activate the cloud services and load them with cloud data. The second phase, migrating from the on-premises solution to the cloud, started in January 2023.

Leumi used data tiering to optimize the architecture. Hot data, or data that has been ingested in recent minutes or hours, will reside in [Amazon Aurora](#) for [PostgreSQL](#), a relational database service that combines the speed and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases. Warm data, which is a few days or several months old, will be stored in [Amazon Redshift](#), which uses SQL to analyze structured and semistructured data.

Cold data, or data that is years old, will be stored in a data lake that Leumi will create using [Amazon Simple Storage Service](#) (Amazon S3), an object storage service that provides industry-leading scalability, data availability, and performance. "With a data lake built on Amazon S3, users in our organization can seamlessly access, and analyze all our data, regardless of where it lives, with security and governance controls in place." "The



amount of data we can retain in Amazon S3 is limitless, whereas we could save only a finite amount of data on premises.”

With an Amazon S3 data lake, Leumi can speed up time to insights by reducing the amount of time its teams spend seeking data and shifting that work to analysis. “On-premises infrastructure is significantly more restricted, but using Amazon S3 empowers us to work quickly and be more agile,” says Levi. “We can think of the solution instead of the process. We’re not constrained anymore—we can just work.”

Although the migration won’t be fully completed until December 2023, the cyber team and a few key Leumi stakeholders have already benefited from enhanced data security supported on Amazon Redshift. Leumi SOC analysts, who monitor the bank’s security 24-7, can perform forensic analysis of 8 months of data, about 100 TB, which is much more historical data than they could analyze on premises. As a result, forensic analysis takes hours rather than days to complete.

The bank also uses AWS security services to protect its infrastructure; for example, it uses [AWS Identity and Access Management](#) (AWS IAM), which securely manages identities and access to AWS services, to enforce role-based access control. It also adopted [AWS Key Management Service](#) (AWS KMS), which is used to create, manage, and control cryptographic keys across an organization’s applications and AWS services.

Because Amazon Redshift stores semistructured and structured data in the same table for processing, SOC analysts can perform a forensic analysis without the assistance of data engineers to extract the required data. Additionally, Amazon Redshift materializes part of the larger table in a view with the relevant subset of the data in a matter of minutes, compared with hours on premises. Moreover, with the data-sharing capabilities of Amazon Redshift, the bank can separate consumers so that a threat researcher doesn’t impact an analyst who is working on a current issue’s forensics.

“Because our SOC analysts have all the details of a cyberattack using Amazon Redshift, they make better decisions, and faster ones, because they don’t need an engineer to work on the data,” says Levi.

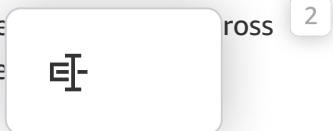
Relying on managed services such as Amazon Redshift means that Leumi’s cyber team doesn’t have to perform updates or patch servers. Leumi can scale an Amazon Redshift cluster in 6 minutes, compared with the months it took to upgrade and expand its on-premises infrastructure. “It saves time and money when data engineers and scientists do what they were hired to do,” says Levi. “We can work in our domain and not deal with errors and malfunctions—and our customers are saying the same about our products.”

Outcome | Helping Customers Do More with Their Data on AWS

Leumi now has access to a wealth of data. And being on AWS, the bank can fully use that data. “We are just getting started with more advanced analytics,” says Levi. “We are going to develop so many use cases that I can’t even imagine where we’ll be in the next year. On AWS, the service is constantly improving, and we continue to evolve alongside it.”

About Bank Leumi

Founded in 1902, Bank Leumi is Israel’s largest bank according to market cap, operating some 1,000 branches across Israel. Leumi is a pioneer in the global digital banking field, offering a wide variety of advanced online banking services.



AWS Services Used

Amazon Redshift

Amazon Redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes, using AWS-designed hardware and machine learning to deliver the best price performance at any scale.

[Learn more »](#)

Amazon Aurora for PostgreSQL

Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication, and integrations with other AWS services.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

AWS Identity and Access Management

With AWS Identity and Access Management (IAM), you can specify who or what can access services and resources in AWS, centrally manage fine-grained permissions, and analyze access to refine permissions across AWS.

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